

Proposal: Dice Boss Fighter

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Idea: The player will fight a monster boss.

The game is in an event controlled loop until the player wins or loses to the boss.

The event being either the boss or players health is equal to 0

ex. `While health > 0 while (playerHealth != 0 || bossHealth !=0)`

The player plays as a hero with 4 attacks.

Player:

The player will have 4 attacks which will be stored in methods in its class.

In the players turn:

The player will choose one of their hero's 4 attacks to use on the boss.

Player attacks will hit or miss depending on the 12 sided dice roll (which will be randomly generated) (a random number between 1 and 12, inclusive).

Example:

Choose your attack

-sword: deals 5 damage (needs to roll 3 or higher) (5 points)

-bow: deals 10 damage (needs to roll 6 or higher) (10 points)

-lightning magic: 15 damage (needs to roll 8 or higher) (20 points)

-fire magic: deals 20 damage (needs to roll 10 or higher) (50 points)

If the player chooses the bow and rolls a 6, then it will hit the boss.

If the player rolls a lower than 6, then the attack misses and the boss takes no damage.

Boss:

The boss will have its own class and its own health bar. The boss has 1 attack (death slash), but the power of its damage will be determined by the 6 sided dice roll.

(rolls a one): attack misses

(rolls a two or three): deals 5 damage

(rolls a four or five): deals 10 damage

(rolls a six): deals 15 damage

After each turn, the program will display the health of the boss and the player.

The boss has 30 health and the player has 25 health.

Scoring system:

Scoring is determined by a point system.

Successfully hitting the boss increases your score, and missing attacks doesn't increase the score.

You can get higher scores by taking risks

Going back to the knight example, if the player chooses the bow, there is a higher chance of missing the attack than if you choose the sword. However, using the bow gives you more points than using the sword.

Basically, if you take a high risk, there is a chance of getting a high reward (a higher score).

Score Board:

After each victory against the boss, the scores will be recorded in an arraylist.

If you lose, the score will not be recorded.

If you choose to access the scoreboard, then the page will display the top 3 scores.

Classes:

- Main.java: where the user inputs actions (Start, Retry, Quit, Scoreboard)
- Entity.java: superclass of Player & Boss, (health)
- Player.java: contains player data (attacks, player health),
- Boss.java: contains boss data (attacks, health)
- ScoreBoard.java: contains an arraylist of scores, calculates top 3.

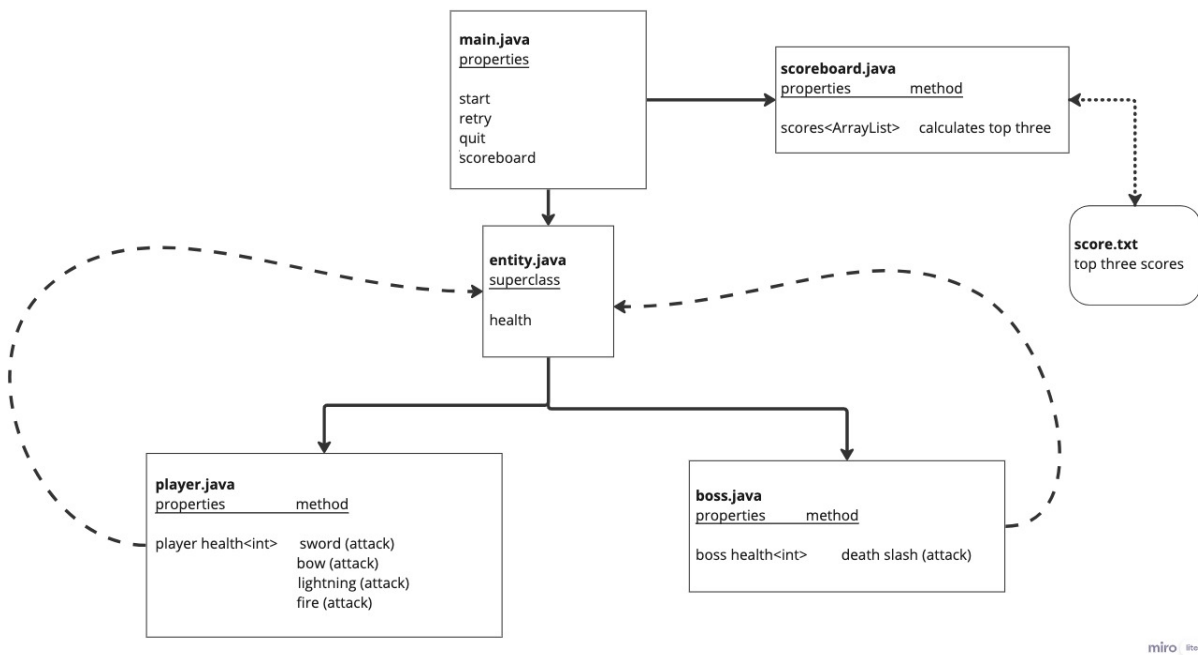
Interfaces:

- Attack choice: choose one of the 4 attacks for the turn (sword, bow, lightning, fire)
- Choose to quit (to main menu) or try again
- Option to see the top 3 scores

Files:

- topThree: contains the top 3 current scores (text file)

UML Diagram:



File Interaction Diagram:

Front End Program:

Welcome!

Start or see Top 3 Scores?

"Start"

Player health: 25

Boss health: 30

Player's turn:

Choose your attack

-sword: deals 5 damage (needs to roll 3 or higher) (5 points)

-bow: deals 10 damage (needs to roll 6 or higher) (10 points)

-lightning magic: 15 damage (needs to roll 8 or higher) (20 points)

-fire magic: deals 20 damage (needs to roll 10 or higher) (50 points)

"bow"

Player uses a bow!

The attack hits the boss for 10 points!

Boss Turn:

Boss uses death slash

The Boss rolled 3!

Player health: 20

Boss Health: 20

Player's turn:

Choose your attack

-sword: deals 5 damage (needs to roll 3 or higher) (5 points)

-bow: deals 10 damage (needs to roll 6 or higher) (10 points)

-lightning magic: 15 damage (needs to roll 8 or higher) (20 points)

-fire magic: deals 20 damage (needs to roll 10 or higher) (50 points)

"lightning"

Player uses lightning!

The attack misses and no points are given.

Boss Turn:

Boss uses death slash

The Boss rolled 4!

Player health: 10

Boss Health: 20

Player's turn:

Choose your attack

-sword: deals 5 damage (needs to roll 3 or higher) (5 points)

-bow: deals 10 damage (needs to roll 6 or higher) (10 points)

-lightning magic: 15 damage (needs to roll 8 or higher) (20 points)

-fire magic: deals 20 damage (needs to roll 10 or higher) (50 points)

"fire"

Player uses fire!

The attack hits for 50 points!

The boss is defeated!

Final Score: 60

You won!

Retry, Quit, or see Top 3 scores?

"Quit"

Welcome!

Start or see Top 3 Scores?

"Scoreboard"

1. 70

2. 40

3. 35

"Quit"

Welcome!

Start or see Top 3 Scores?